

Abdul Wahid

Lab Report

OSINT Footprinting – Maltego CE

Executive Summary

This report documents OSINT (Open Source Intelligence) gathering performed on the target domain certifiedhacker.com using Maltego CE (Community Edition) on Kali Linux. The objective was to collect publicly available information about the target organization's digital infrastructure without direct interaction with the target systems.

Maltego CE was used to run a series of transforms on the target website entity (www.certifiedhacker.com). A total of 12 successful reconnaissance steps were completed, gathering information including domain structure, DNS records, mail and name servers, IP address, geographical location, WHOIS ownership data, email addresses, phone numbers, external link relationships, and historical website snapshots via the Wayback Machine.

Two transforms — Server-Side Technology (BuiltWith) and Query Open Ports — were attempted but returned no results due to limitations of the free Community Edition. These have been noted and documented with their expected functionality for reference.

Key findings include:

- Domain hosted on Bluehost servers (ns1.bluehost.com, ns2.bluehost.com)
- Server IP: 162.241.216.11 — geographically located in Phoenix, United States
- 7 subdomains identified including mail, ftp, smtp, webmail, imap, pop, news
- Domain registered under Perfect Privacy LLC via Network Solutions LLC registrar
- 2 phone numbers extracted from WHOIS: +1 570 708 8622 and +1 877 722 8662
- 25+ historical website snapshots found (oldest: 2004 Jul 30)
- External links to Google Tag Manager and w3.org reveal third-party service dependencies

All gathered intelligence can be used by a penetration tester or threat actor to map the organization's attack surface, identify vulnerable entry points, and plan social engineering or technical exploitation strategies.

Lab Setup & Configuration

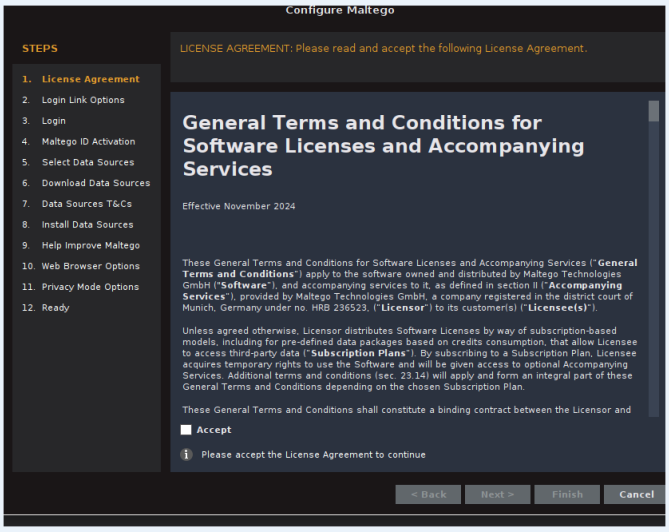
Step 1 — Launch Maltego CE on Kali Linux

- Launched Maltego Graph (Desktop) 4.11.2 from Kali Linux application menu
- Accepted the License Agreement on first launch



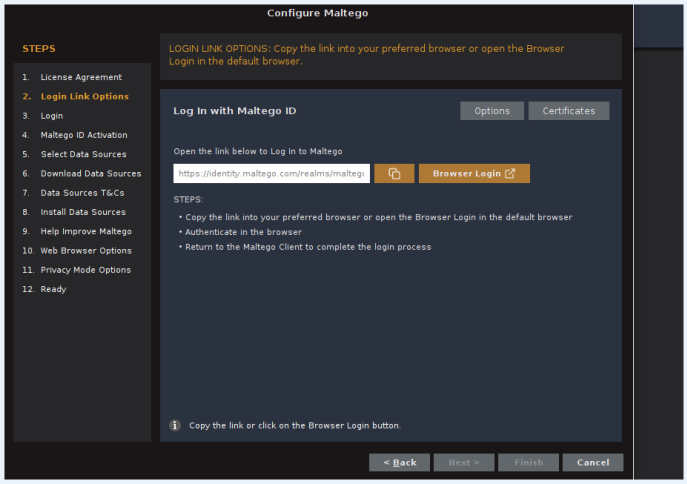
Step 2 — License Agreement

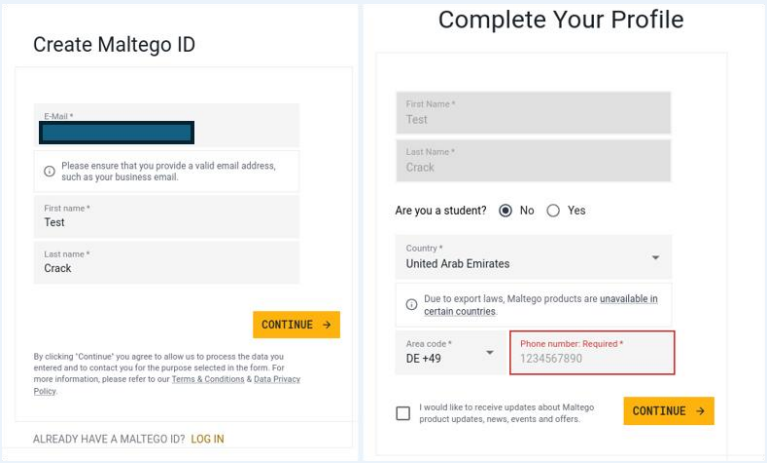
- Read and accepted the General Terms and Conditions for Software Licenses
- Clicked Accept checkbox and proceeded to Login Link Options



Step 3 — Login & Account Setup

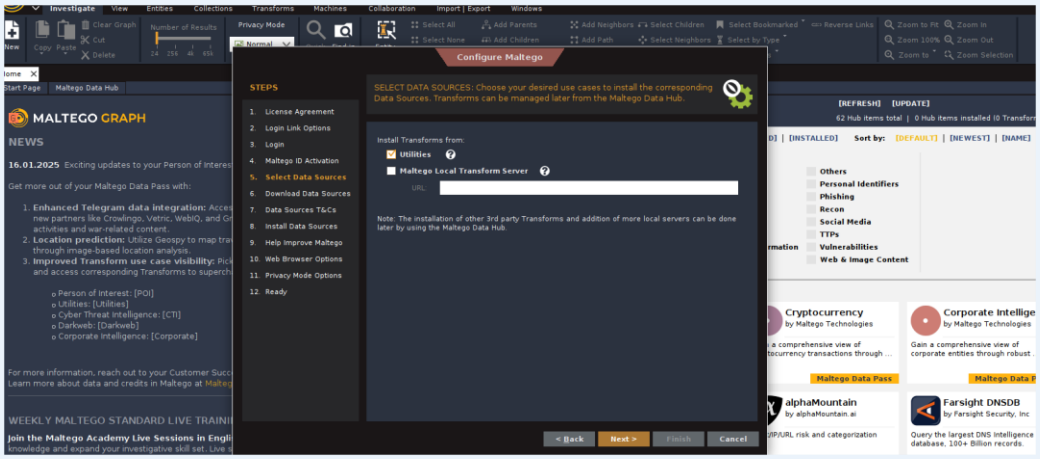
- Used Browser Login to authenticate with Maltego ID
- Created a Maltego account using student email: 243699@students.au.edu.pk
- Completed profile registration (Name: Test Crack, Country: United Arab Emirates)





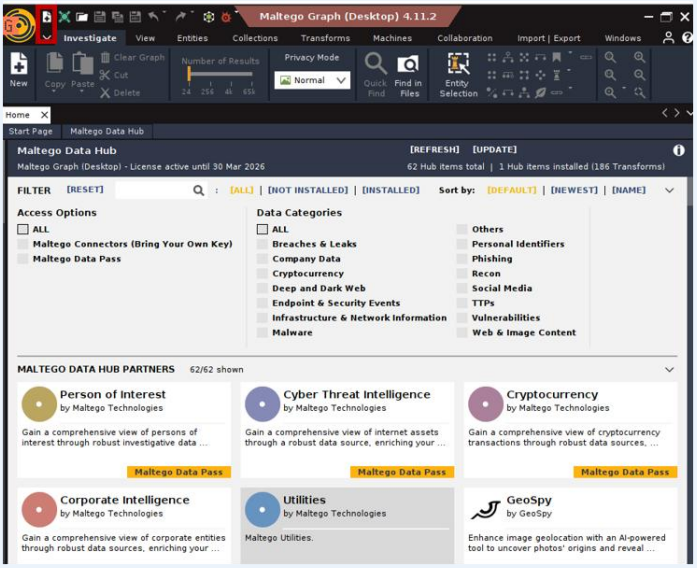
Step 4 — Select Data Sources & Install Transforms

- In the Configure Maltego wizard, selected Utilities as the data source
- Maltego Data Hub displayed 62 total hub items with 186 transforms available
- Installed default Utilities transforms required for the lab



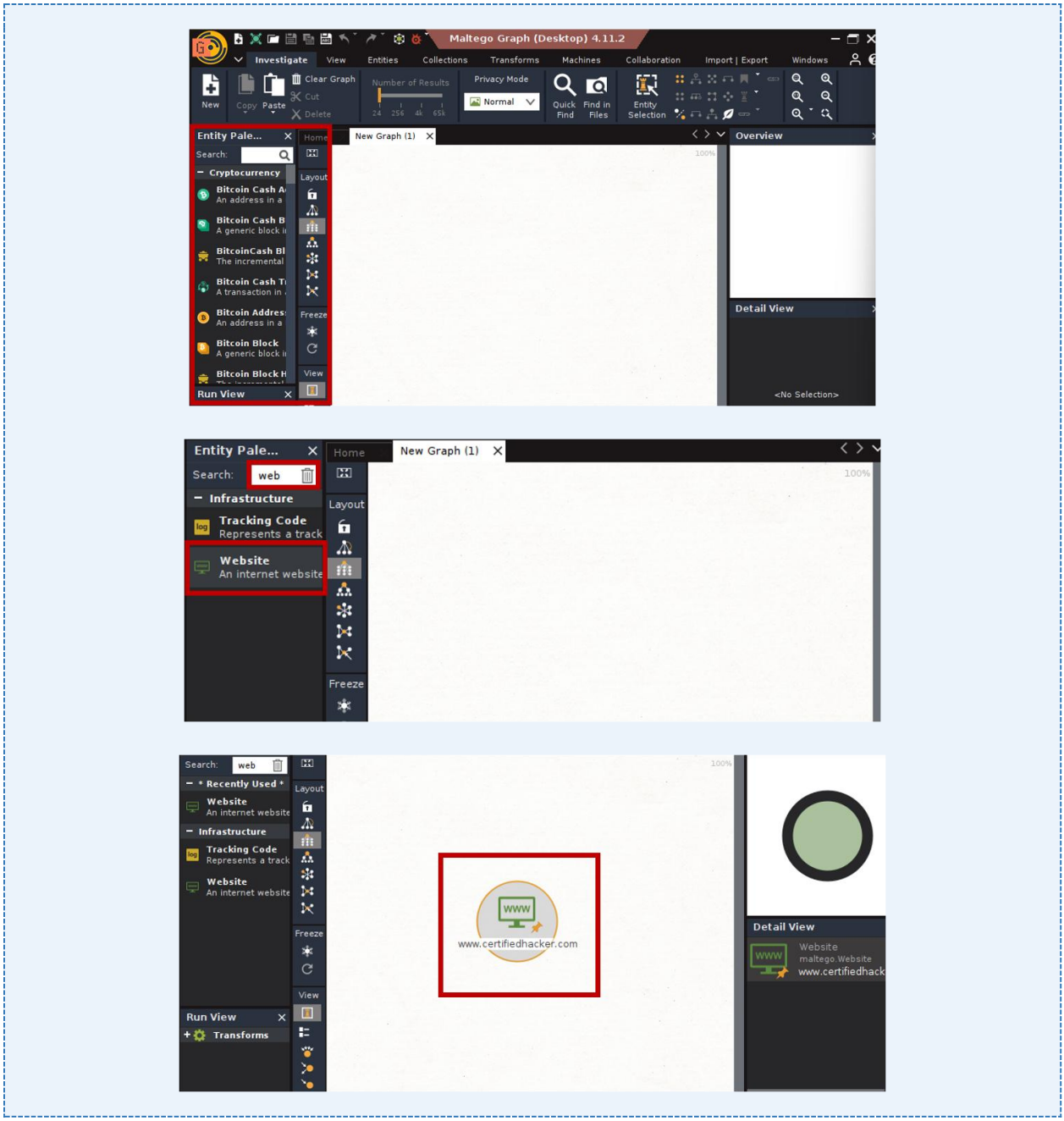
Step 5 — Create New Graph

- From the Maltego home screen, clicked New to create a blank graph
- New Graph (1) was created and opened in the graph workspace



Step 6 — Add Website Entity

- In the Entity Palette (left pane), searched for 'Website' in the search box
- Dragged and dropped the Website entity (under Infrastructure) onto the blank graph
- Double-clicked the entity and renamed it to: www.certifiedhacker.com
- Pressed Enter to confirm the entity name



Task 1 — OSINT Gathering on certifiedhacker.com

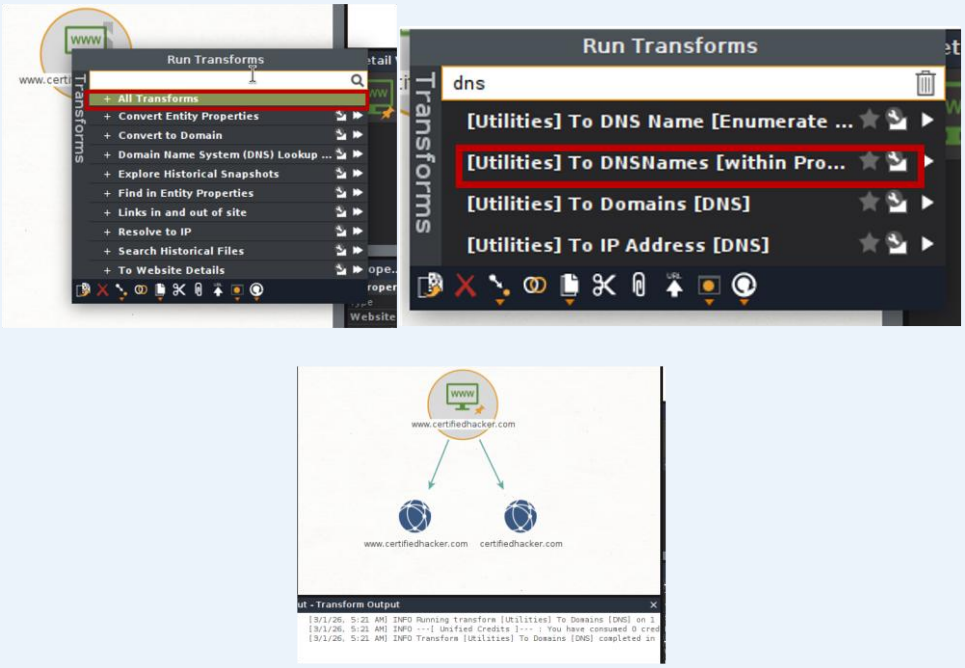
Step 7 — Identify Domain

Procedure:

- Right-clicked on www.certifiedhacker.com entity → All Transforms
- In the search box typed: DNS
- Selected: [Utilities] To Domains [DNS]

Result:

- Transform successfully identified 2 domain entities linked to the website
- **Domain 1:** Domains found:
- www.certifiedhacker.com
- certifiedhacker.com
- These domains confirm the DNS structure of the target and can be used to investigate subdomains, mail servers, and associated IP addresses



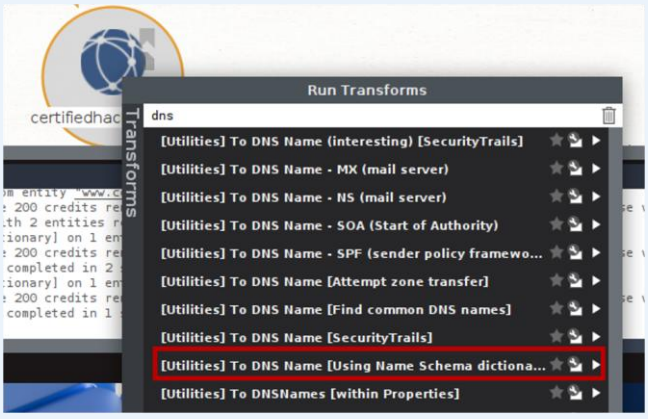
Step 8 — Identify Domain Name Schema (Subdomains)

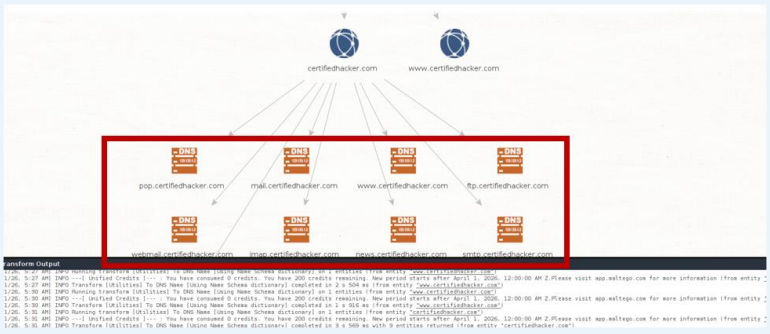
Procedure:

- Right-clicked on certifiedhacker.com domain entity → All Transforms
- Searched for: DNS
- Selected: [Utilities] To DNS Name [Using Name Schema dictionary]

Result:

- Transform returned 9 entities — multiple active subdomains of the target
- Identified subdomains:
- pop.certifiedhacker.com
- mail.certifiedhacker.com
- ftp.certifiedhacker.com
- webmail.certifiedhacker.com
- imap.certifiedhacker.com
- news.certifiedhacker.com
- smtp.certifiedhacker.com
- These subdomains reveal the internal service structure of the organization
- Attackers can use brute force or dictionary attacks on ftp and smtp subdomains to gain unauthorized access





Step 9 — Identify Server-Side Technology

Procedure:

- Right-clicked on www.certifiedhacker.com → All Transforms
- Searched for 'server' and 'tech' in the transform search box

> ⚠ Transform 'To Server Technologies [Using BuiltWith]' is NOT available in Maltego CE (free version). Both searches returned 'No Transforms'.

Expected Result (Paid Version):

- Would identify server-side technologies such as jQuery, ASP.NET, IIS 8, SPF, and other web frameworks
- These technologies can be researched for known CVEs and exploited accordingly

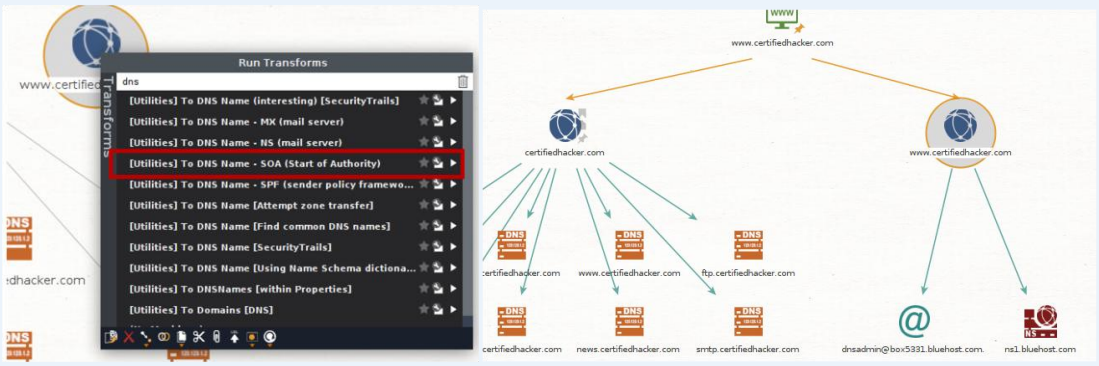
Step 10 — Identify SOA (Start of Authority)

Procedure:

- Right-clicked on certifiedhacker.com entity → All Transforms
- Searched for: DNS
- Selected: [Utilities] To DNS Name - SOA (Start of Authority)

Result:

- Transform successfully returned 2 entities
- **ns1.bluehost.com** Primary Name Server:
- **dnsadmin@box5331.bluehost.com** Domain Admin Email:
- This confirms the domain is hosted on Bluehost servers
- Attackers can exploit vulnerabilities in Bluehost services or use the admin email for targeted phishing attacks



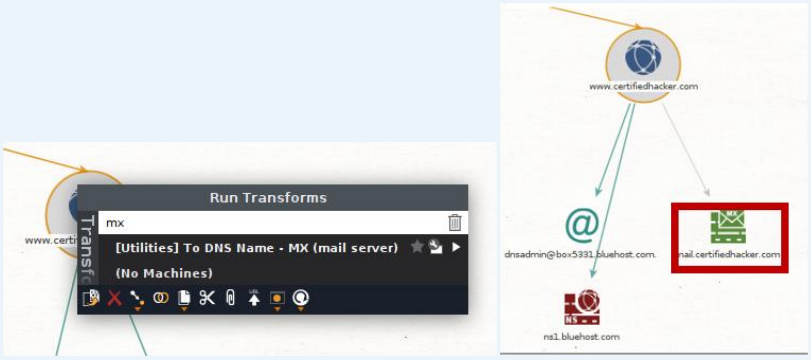
Step 11 — Identify Mail Exchanger (MX Record)

Procedure:

- Right-clicked on certifiedhacker.com domain entity → All Transforms
- Searched for: MX
- Selected: [Utilities] To DNS Name - MX (mail server)

Result:

- Transform executed successfully on certifiedhacker.com
- **mail.certifiedhacker.com** Mail server identified:
- Confirms the domain has a dedicated mail server handling all email communications
- Attackers can exploit this mail server to send spoofed/spam emails or conduct phishing attacks against the organization



Step 12 — Identify Name Servers (NS Record)

Procedure:

- Right-clicked on certifiedhacker.com entity → All Transforms
- Searched for: NS
- Selected: [Utilities] To DNS Name - NS (name server)

Result:

- Transform returned 2 name server entities
- **ns1.bluehost.com** Name Server 1:
- **ns2.bluehost.com** Name Server 2:
- Both name servers confirm the domain is entirely managed through Bluehost infrastructure
- Attackers can exploit these name servers to perform DNS hijacking and URL redirection attacks



Step 13 — Identify IP Address

Procedure:

- Right-clicked on www.certifiedhacker.com website entity → All Transforms
- Searched for: IP
- Selected: [Utilities] To IP Address [DNS]

Result:

- Transform successfully resolved the IP address of the target website
- **162.241.216.11** IP Address identified:
- This is the actual server IP hosting www.certifiedhacker.com
- This IP can be used for port scanning, network enumeration, and vulnerability assessment in subsequent attack phases



Step 14 — Identify Geographical Location

Procedure:

- Right-clicked on IP entity 162.241.216.11 → All Transforms
- Searched for: location
- Selected: [Utilities] To Location [city, country]

Result:

- Transform successfully returned geographical location of the server
- **Phoenix, United States** Location:
- This confirms the target server is physically located in Phoenix, Arizona, USA
- Geographical data assists attackers in performing targeted vishing (voice phishing) or social engineering attacks by impersonating local entities



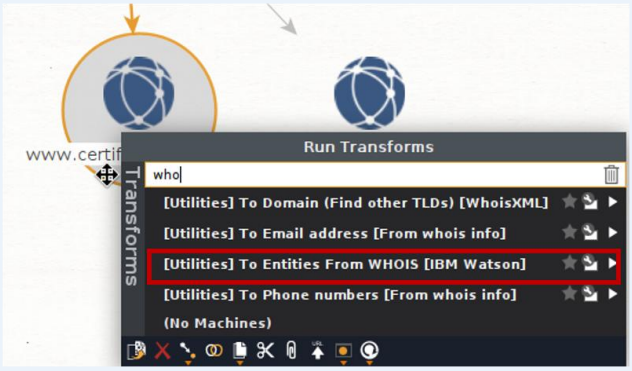
Step 15 — Identify WHOIS Entities

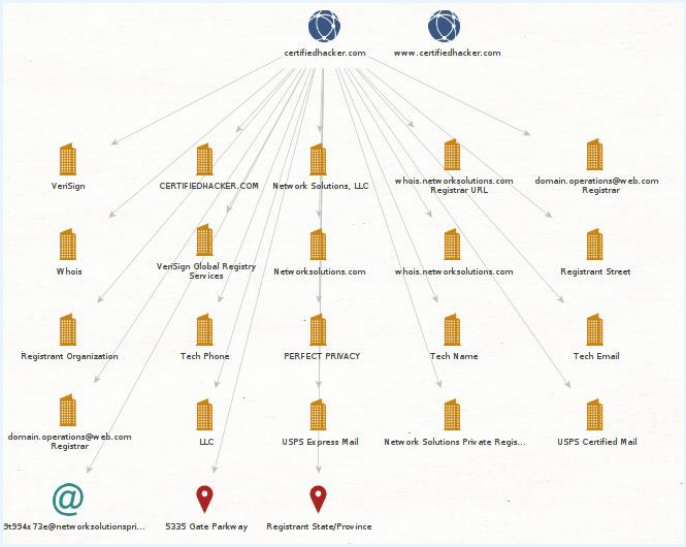
Procedure:

- Right-clicked on certifiedhacker.com domain entity → All Transforms
- Searched for: whois
- Selected: [Utilities] To Entities From WHOIS [IBM Watson]

Result:

- Transform returned complete domain ownership and registration details
- **Network Solutions LLC** Registrar:
- **Perfect Privacy LLC** Registrant Organization:
- **5335 Gate Parkway** Registrant Address:
- **whois.networksolutions.com** Registrar URL:
- Additional entities: Tech Name, Tech Email, Tech Phone, Registrant State/Province
- This WHOIS data can be used to design highly targeted spear-phishing attacks and brute force attempts against admin accounts





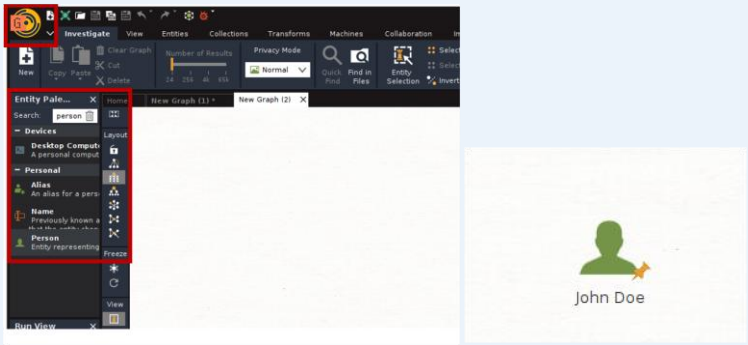
Step 16 — Identify Email Addresses

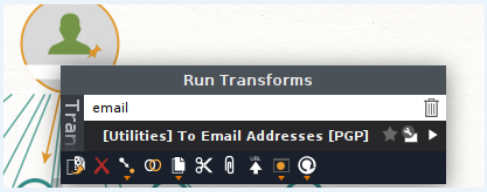
Procedure:

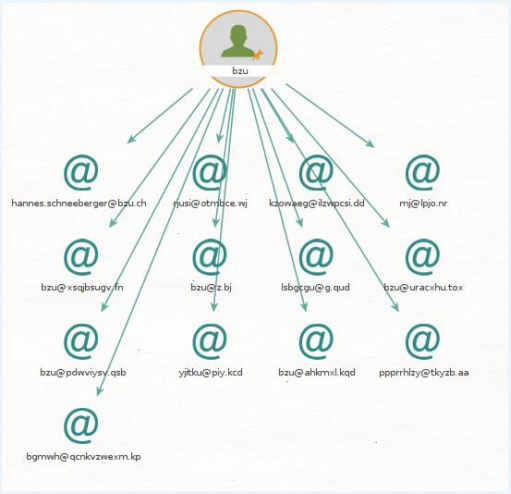
- Created New Graph (2) with a Person entity from Entity Palette → Personal → Person
- Renamed entity to target name: bzu
- Right-clicked on entity → All Transforms → Searched: email
- Selected: [Utilities] To Email Addresses [PGP]

Result:

- Transform successfully returned multiple email addresses associated with the name 'bzu'
- Identified emails include: hannes.schneeberger@bzu.ch, bzu@xsqjbsugv.fn, bzu@z.bj, bzu@uracxhu.tox, and several others
- Analyst must verify which email address actually belongs to the target person
- These email addresses can be used for phishing attacks and credential brute forcing







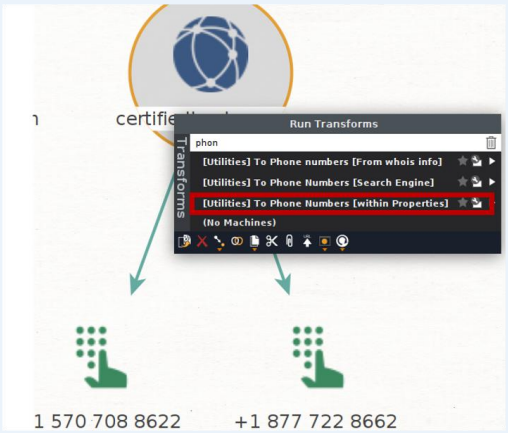
Extra Step 1 — Identify Phone Numbers (WHOIS)

Procedure:

- Right-clicked on certifiedhacker.com domain entity → All Transforms
- Searched for: phone
- Selected: [Utilities] To Phone Numbers [From whois info]

Result:

- Transform executed successfully and returned 2 phone numbers from the WHOIS record
- **+1 570 708 8622** Phone Number 1:
- **+1 877 722 8662** Phone Number 2:
- These numbers belong to the domain registrant or technical contact of certifiedhacker.com
- Attackers can use these numbers to perform vishing (voice phishing) or social engineering attacks to extract sensitive credentials



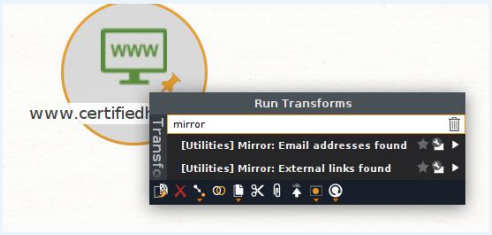
Extra Step 2 — Mirror External Links

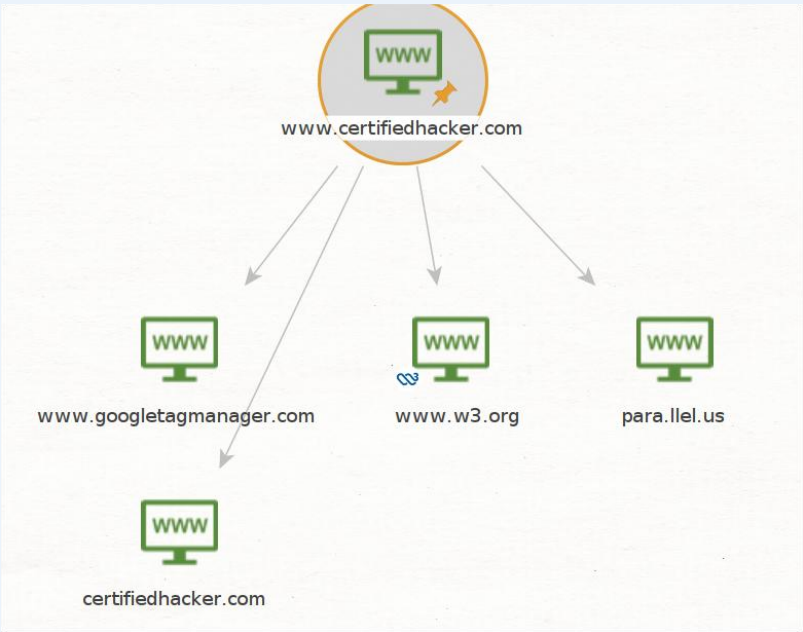
Procedure:

- Right-clicked on www.certifiedhacker.com entity → All Transforms
- Searched for: mirror
- Selected: [Utilities] Mirror: External links found

Result:

- Transform returned 4 external websites linked from the target website
- **www.googletagmanager.com** External link 1:
- **www.w3.org** External link 2:
- **para.llel.us** External link 3:
- **certifiedhacker.com** External link 4:
- These external links reveal third-party services and tracking tools used by the organization
- Attackers can target these linked domains to perform supply chain attacks against the organization





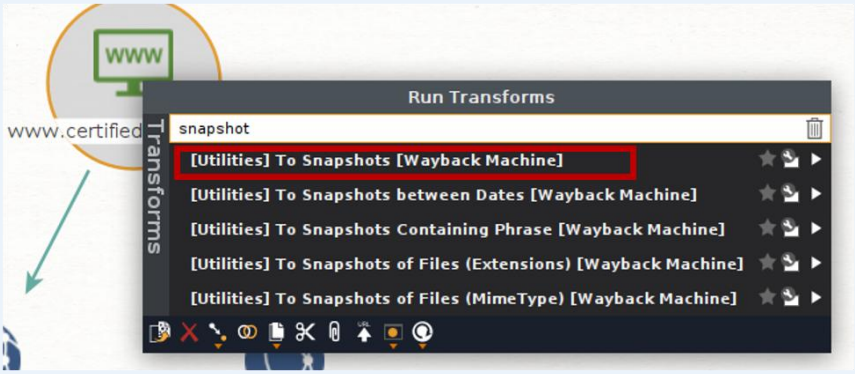
Extra Step 3 — Wayback Machine Historical Snapshots

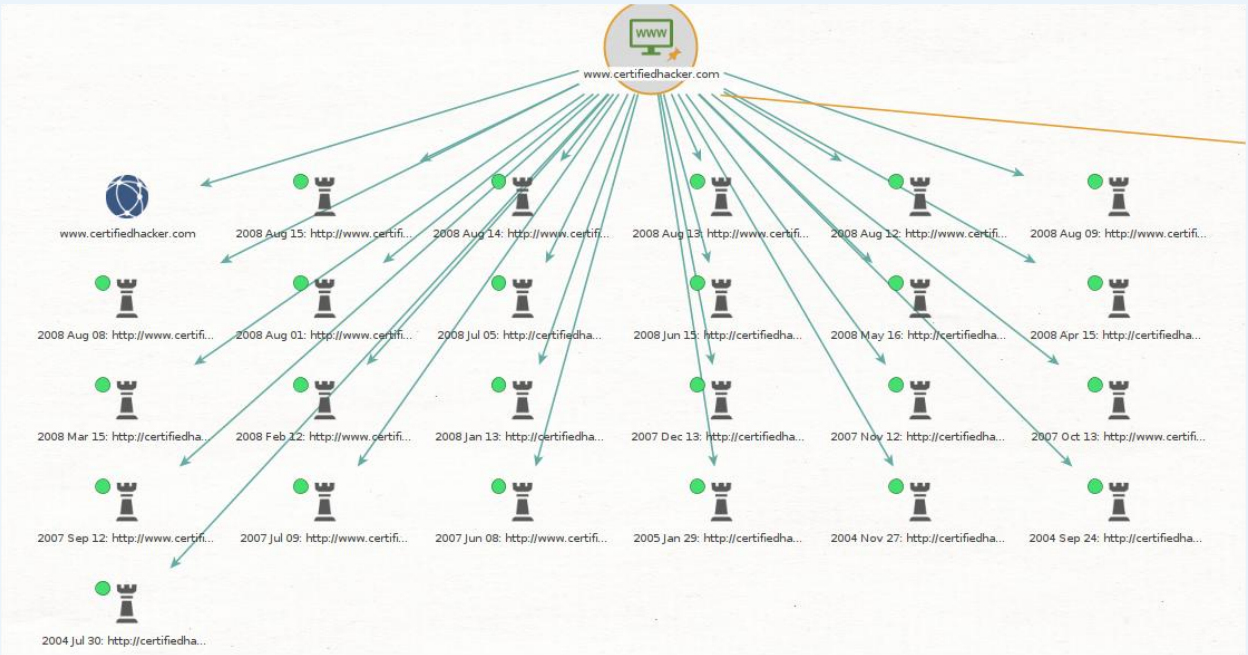
Procedure:

- Right-clicked on www.certifiedhacker.com entity → All Transforms
- Searched for: snapshot
- Selected: [Utilities] To Snapshots [Wayback Machine]

Result:

- Transform successfully returned 25+ historical snapshots of the target website
- **2004 to 2008** Date range:
- **2004 Jul 30 — http://certifiedhacker.com** Oldest snapshot:
- Historical snapshots can reveal old login pages, hidden directories, and outdated software versions no longer visible on the live site
- Attackers can analyze old website versions to identify vulnerabilities, deprecated configurations, or exposed files that may still be accessible on the server





Extra Step 4 — Query Open Ports

Procedure:

- Right-clicked on www.certifiedhacker.com entity → All Transforms
- Searched for: port
- Selected: [Utilities] To Web site [Query ports]

> ⚠ Transform was available but returned no results — this is a free version limitation of Maltego CE.

Expected Result (Paid Version):

- Would identify open ports such as 80 (HTTP), 443 (HTTPS), 22 (SSH), 21 (FTP) on the target server
- Open port information enables an attacker to identify running services and launch targeted exploitation of specific protocols

Summary of All Gathered Intelligence

#	Step	Transform Used	Key Finding	Status
1	Domain Identification	To Domains [DNS]	www.certifiedhacker.com, certifiedhacker.com	☑ Complete
2	Domain Name Schema	To DNS Name [Schema dictionary]	7 subdomains: mail, ftp, smtp, pop, imap, webmail, news	☑ Complete
3	Server-Side Technology	To Server Technologies [BuiltWith]	Not available in CE version	⚠ CE Limit
4	SOA Record	To DNS Name - SOA	ns1.bluehost.com | dnsadmin@box5331.bluehost.com	☑ Complete
5	MX Record	To DNS Name - MX	mail.certifiedhacker.com	☑ Complete
6	NS Record	To DNS Name - NS	ns1.bluehost.com, ns2.bluehost.com	☑ Complete
7	IP Address	To IP Address [DNS]	162.241.216.11	☑ Complete
8	Geographical Location	To Location [city, country]	Phoenix, United States	☑ Complete
9	WHOIS Entities	To Entities From WHOIS [IBM Watson]	Registrar: Network Solutions LLC | Org: Perfect Privacy LLC	☑ Complete
10	Email Addresses	To Email Addresses [PGP]	hannes.schneeberger@bzu.ch + several others	☑ Complete
11	Phone Numbers	To Phone Numbers [From whois info]	+1 570 708 8622 | +1 877 722 8662	☑ Complete
12	Mirror External Links	Mirror: External links found	googletagmanager.com, w3.org, para.llel.us	☑ Complete
13	Wayback Machine	To Snapshots [Wayback Machine]	25+ snapshots from 2004–2008	☑ Complete
14	Query Open Ports	To Web site [Query ports]	No results — CE version limitation	⚠ CE Limit

Conclusion

This lab successfully demonstrated the use of Maltego CE as an OSINT foot printing tool. Using only passive, non-intrusive transformations on publicly available data sources, a significant amount of intelligence about the target organization's digital infrastructure was gathered.

The information collected — including DNS records, server location, ownership data, email addresses, phone numbers, and historical website data — forms a comprehensive threat profile that could be used in the reconnaissance phase of a penetration test or ethical hacking engagement.

Two transforms were unavailable due to the Maltego CE (free) version limitation: Server-Side Technology detection and Open Port querying. These would provide additional attack surface information in a professional or paid edition.